

Online Appendix

Companion to “可验证性与首次生成式人工智能披露的市场反应分化：投资者推断异质性视角。”

This appendix documents the large language model (LLM) investor-persona elicitation protocol underlying Section 7.2 and Tables 23–24, including verbatim prompts, model versions, and API parameters.

Appendix A. LLM investor-persona elicitation protocol

A.1 Design

For each first-disclosing firm ($n = 735$), the firm's strict-main GenAI disclosure segment was submitted to two cross-family frontier LLMs through three independent, single-shot elicitations: (T1) a verifiability classification with a 0–3 score; (T2) a sophisticated/institutional-investor estimate of the probability of genuine, substantive GenAI adoption; and (T3) a retail-investor estimate of the same. The cognitive separation index is $CSI = P(\text{retail} | \text{text}) - P(\text{sophisticated} | \text{text}) = T3 - T2$. Each call was independent and blind: the model received only the construct definition (T1) or persona instruction (T2/T3) and the raw Chinese MD&A segment (truncated to 1,800 characters); it had no access to the keyword dictionary, the human codings, or the other model's outputs.

Models and access. (i) GPT-5.5 (OpenAI-family); (ii) Anthropic Claude Opus 4.6 (model identifier claude-opus-4-6). Both were queried via API on 2026-05-20 at temperature = 0, max_tokens = 1024 (Anthropic API version 2023-06-01). Calls were issued per segment, independently, with deterministic decoding (temperature 0); each model returned a strict JSON object that was parsed programmatically. The gold-standard reliability check (Table 4, Table 20) used the same protocol on 180 stratified segments blind to the algorithm labels.

A.2 Prompts (verbatim)

System prompt — T1 (verifiability annotation):

你是独立文本标注员。仅依据片段文本本身、按下面定义做三分类，不使用关键词清单。

可验证(**verifiable**)=含可被外部观察的实施证据：已部署/已上线/明确应用场景/可识别合作伙伴/可衡量推广/明确时间锚定/命名具体工具或产品。

软实质性(**soft_substantive**)=有应用导向语言但缺乏上述可验证实施证据。

象征性(**symbolic**)=仅宽泛、愿景式或概念性表述，或与 GenAI 无关的财报样板。

score: 0=symbolic, 1=soft_substantive, 2=verifiable 但弱, 3=verifiable 且充分。

只输出 JSON: {"category": "verifiable|soft_substantive|symbolic", "score": 0-3}

[Translation] You are an independent text annotator. Classify the segment into three types based solely on the segment text, without any keyword list. Verifiable = contains externally observable implementation evidence (deployed / launched / explicit use case / identifiable partner / measurable rollout / explicit time anchor / named specific tool or product). Soft substantive = application-oriented language but lacking such verifiable evidence. Symbolic = only broad, visionary, or conceptual statements, or GenAI-irrelevant boilerplate. score: 0 = symbolic, 1 = soft_substantive, 2 = verifiable but weak, 3 = verifiable and sufficient. Output only JSON: {"category": "...", "score": 0-3}.

System prompt — T2 (sophisticated / institutional investor):

你是资深机构投资者/卖方资深分析师，擅长穿透年报措辞，对缺乏可被外部核实的实施证据的表述系统性打折扣。仅依据下面给定的中文年报披露片段本身，估计该公司已经真实、实质性落地采用生成式人工智能的概率。

只输出 JSON: {"p": 0 到 1 之间的一个小数}，不要任何解释。

[Translation] You are a senior institutional investor / senior sell-side analyst, skilled at seeing through annual-report wording and systematically discounting statements that lack externally verifiable implementation evidence. Based solely on the given Chinese annual-report disclosure segment, estimate the probability that the firm has genuinely and substantively adopted generative AI. Output only JSON: {"p": a decimal in [0,1]}, no explanation.

System prompt — T3 (retail investor):

你是普通个人投资者（散户），信息处理时间与专业能力有限，主要凭披露读起来给你的总体印象快速判断，不做深入尽职调查。仅依据下面给定的中文年报披露片段本身，给出你认为该公司已经真实采用生成式人工智能的程度。

只输出 JSON: {"p": 0 到 1 之间的一个小数}，不要任何解释。

[Translation] You are an ordinary individual (retail) investor with limited processing time and expertise, judging mainly by the overall impression the disclosure gives, without deep due diligence. Based solely on the given Chinese segment, give the degree to which you believe the firm has genuinely adopted generative AI. Output only JSON: {"p": a decimal in [0,1]}, no explanation.

User message (all three tasks): 披露片段: \n<segment> – i.e., the literal prefix “披露片段: ” followed by the raw MD&A segment, truncated to 1,800 characters.

A.3 Output and aggregation

Each response was a strict JSON object parsed across OpenAI- and Anthropic-style response envelopes. Per-firm T1/T2/T3 values were aggregated to the firm level; the verifiability score validates against the strict-main keyword classification (Spearman $\rho = 0.435$ for GPT-5.5 and 0.387 for Claude Opus 4.6, Table 24), and CSI is strictly positive and rises with verifiability (Table 24).

Appendix B. Human coding reliability

Two independent coders blind-encoded 180 stratified anonymized segments (60 per disclosure type) using only the construct definition (Section 3.3), with no access to the algorithm labels: overall agreement 96.7% (174/180), Cohen's $\kappa = 0.949$. The same 180 gold-standard segments were independently blind-classified by the two LLMs ($\kappa = 0.70$ and 0.54 versus the human consensus; $\kappa = 0.68$ between models), as reported in Section 6.4.

Appendix C. Identification diagnostics

Oster (2019) coefficient-stability bounds, the inverse-probability-weighted DiD covariate-balance diagnostics, the Sun and Abraham (2021) interaction-weighted estimates, and the Callaway–Sant'Anna-style group-time ATT are summarized in the main text (Tables 7, 14, 18, 19). Full per-

covariate weighted balance statistics and the complete elicitation output files are available from the authors on request.